yEvo Mutation Browser

The yEvo Mutation Browser is a web tool that enables users to visualize evolved strain mutations in yEvo data at both the gene and chromosome levels, compare findings under various conditions, and upload their own data, benefiting both educators and the yeast genetics community.

Installation

The easiest method to use the mutation browser is to access it via the web here

To download the code, install the necessary packages, run the script, and edit the master files, please see below for step-by-step instructions

Interface Design

The yEvo Mutation browser contains two sections: the filtering column on the left and the visualizations on the right. Upon first loading, the app displays mutation data across the entire yEvo dataset. The user has the option to 1) change what data is being visualized (filtering) and 2) change the visualization method. Both options are detailed below.

Filtering

By default, the mutation browser displays all data. Users can filter this data in two ways: by student class or by selection condition.

As shown below, when selected, dropdowns appear corresponding to either option.

```
View by Class
    Instructor Dropdown
    Year Dropdown
    Sample Name Dropdown
View by Condition
     Condition Dropdown
```

Below is an example of filtering on the Chromosome Map

Uploading Data (Temporarily)

- Default Data: The mutation browser is preloaded with all mutation data collected from yEvo experiments since 2018. Users can explore this dataset, or upload their own temporarily.
- Data Format: To upload, users must have their mutation data (in .csv form) in the following format:

After uploading, the browser will ask for additional information (instructor and year)

** TODO: don't we have helper tools to get this in the right format?

Note: you can also permanently edit the master file such that your data will always appear in the browser. To do this, more instructions will be given below here (you also have to run the browser through locally/through the code installation)

Visualization Methods

- Chromosome Map
- Variant Pie Chart
- SNP Counts
- Gene View

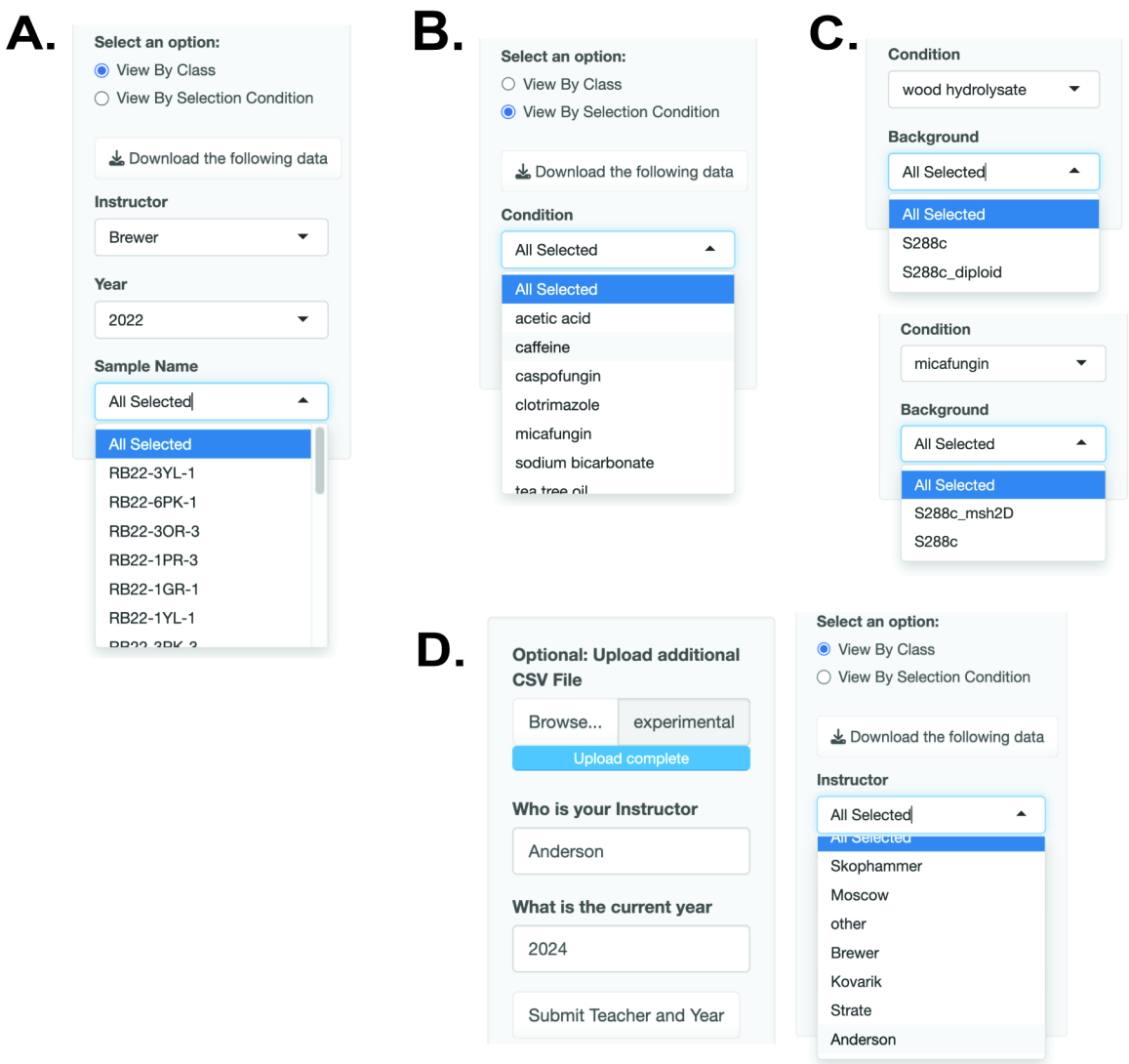

Figure 1: Filtering side panel

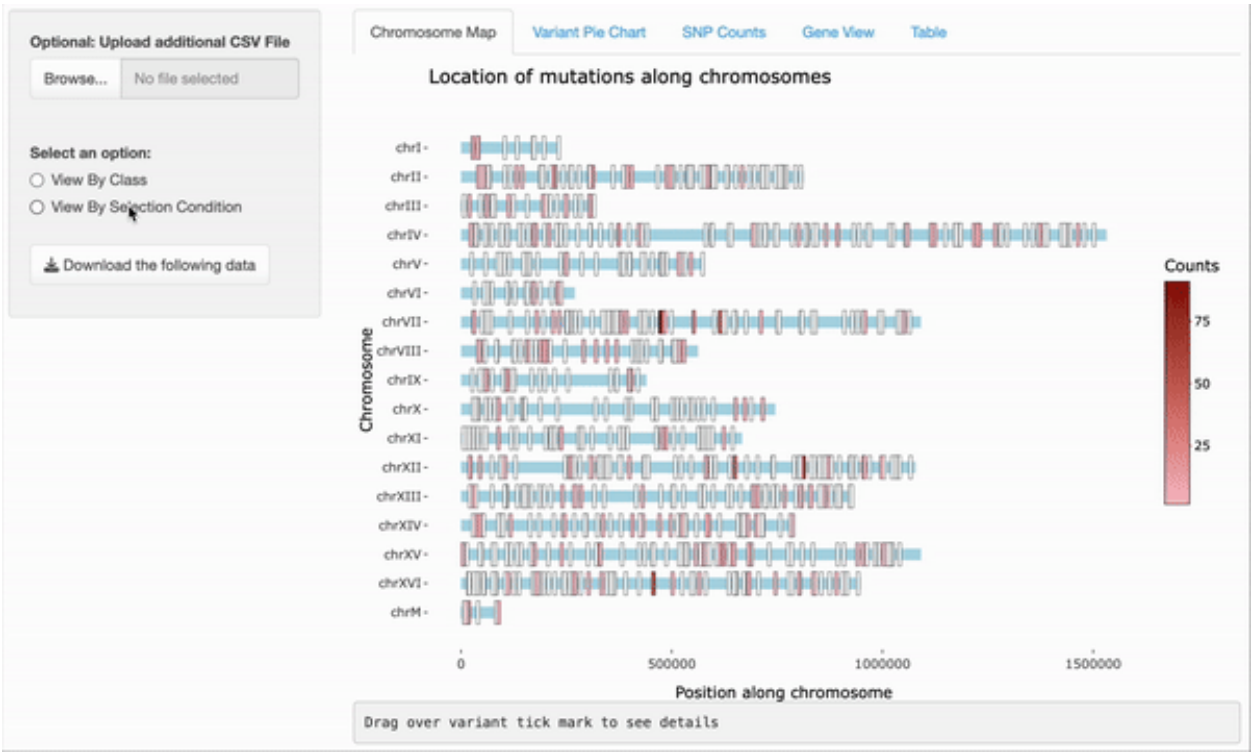

Figure 2: Chromosome map plot

CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	ANNOTATION	REGION	GENE	PROTEIN	seq_file	background	condition	instructor	year	sample
chrII	724851	.	C	G	225	PASS	DP=28;VDB=0.	coding-nonsynon	YBR255W	MTC4	Y132*	1_G4_S12_samt	S288c	clotrimazole	Jasmine	2023	JS23HACK
chrIX	195641	.	C	A	225	PASS	DP=39;VDB=0.	coding-nonsynon	YIL089W	YIL089W	Q15K	1_G4_S12_samt	S288c	clotrimazole	Jasmine	2023	JS23HACK
chrM	85640	.	CCTA	CCTATACTA	222	PASS	INDEL;IDV=155	coding-nonsynon	Q0297	Q0297	S29indel	1_G4_S12_samt	S288c	clotrimazole	Jasmine	2023	JS23HACK
chrVII	471400	.	G	A	225	PASS	DP=46;VDB=0.	coding-nonsynon	YGL013C	PDR1	T300I	1_G4_S12_samt	S288c	clotrimazole	Jasmine	2023	JS23HACK
chrXV	586868	.	G	C	225	PASS	DP=27;VDB=0.	5'-upstream	YOR140W	SFL1	NA	1_G4_S12_samt	S288c	clotrimazole	Jasmine	2023	JS23HACK
chrXVI	452468	.	T	G	225	PASS	DP=53;VDB=0.	coding-nonsynon	YPL057C	SUR 1.00	Y197S	1_G4_S12_samt	S288c	clotrimazole	Jasmine	2023	JS23HACK
chrIV	439359	.	G	T	225	PASS	DP=11;VDB=0.	coding-nonsynon	YDL007W	RPT2	*438L	4_A2_S19_samt	S288c	clotrimazole	Jasmine	2023	JS23HACK
chrIV	1012249	.	A	C	225	PASS	DP=15;VDB=0.	5'-upstream	YDR275W	BSC2	NA	4_A2_S19_samt	S288c	clotrimazole	Jasmine	2023	JS23HACK
chrVII	611152	.	T	C	228	PASS	DP=18;VDB=0.	coding-nonsynon	YGR060W	ERG25	S197P	4_A2_S19_samt	S288c	clotrimazole	Jasmine	2023	JS23HACK

Figure 3: CSV structure

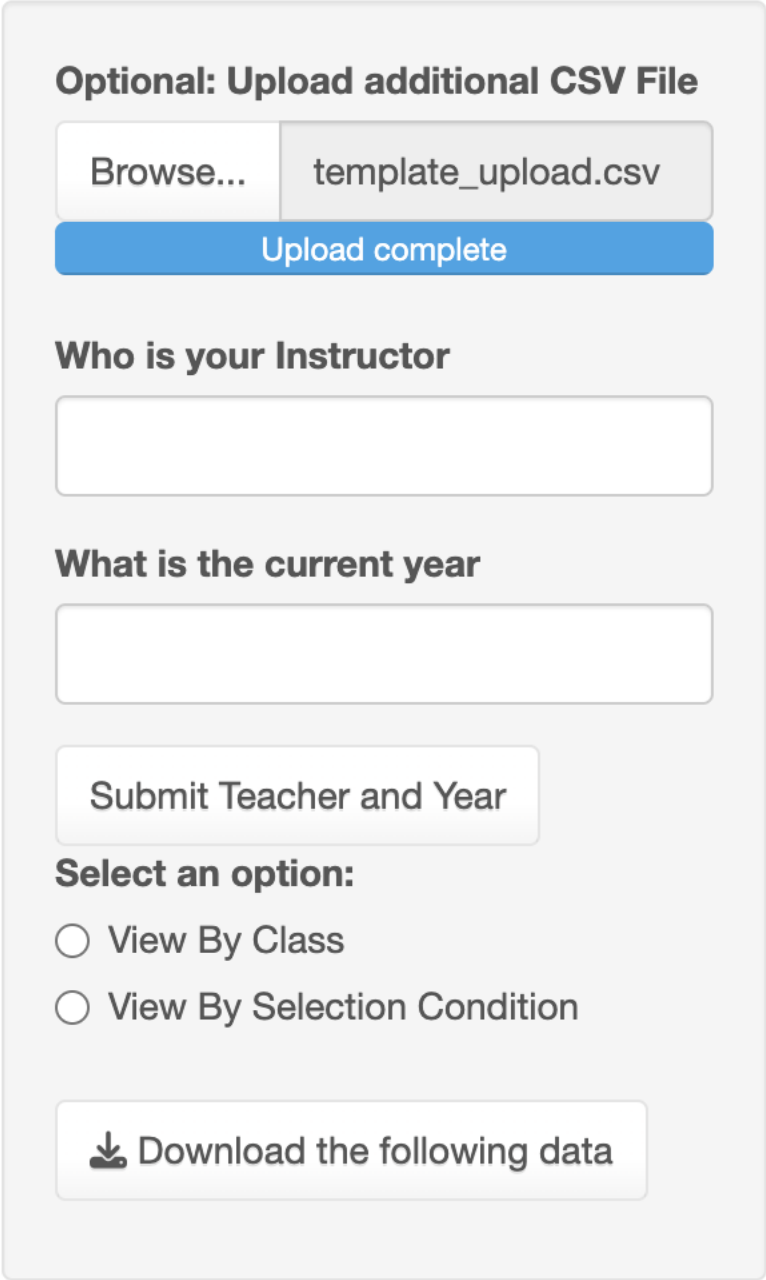

Figure 4: CSV upload

Chromosome Map

The Chromosome Map is a linear representation of the 16 nuclear chromosomes along with the mitochondrial genome of Saccharomyces cerevisiae. This plot displays the locations of all mutated genes in the selected dataset, marked at their chromosomal positions. Tick marks along the linear chromosomes indicate these sites, and hovering the mouse over each tick mark reveals the name of the mutated gene, as well as how many times it was mutated in repeat experiments. Users can also zoom in on specific regions by clicking and dragging, offering a closer look at mutations in a given area.

Default (Unfiltered) Chromosome Map:

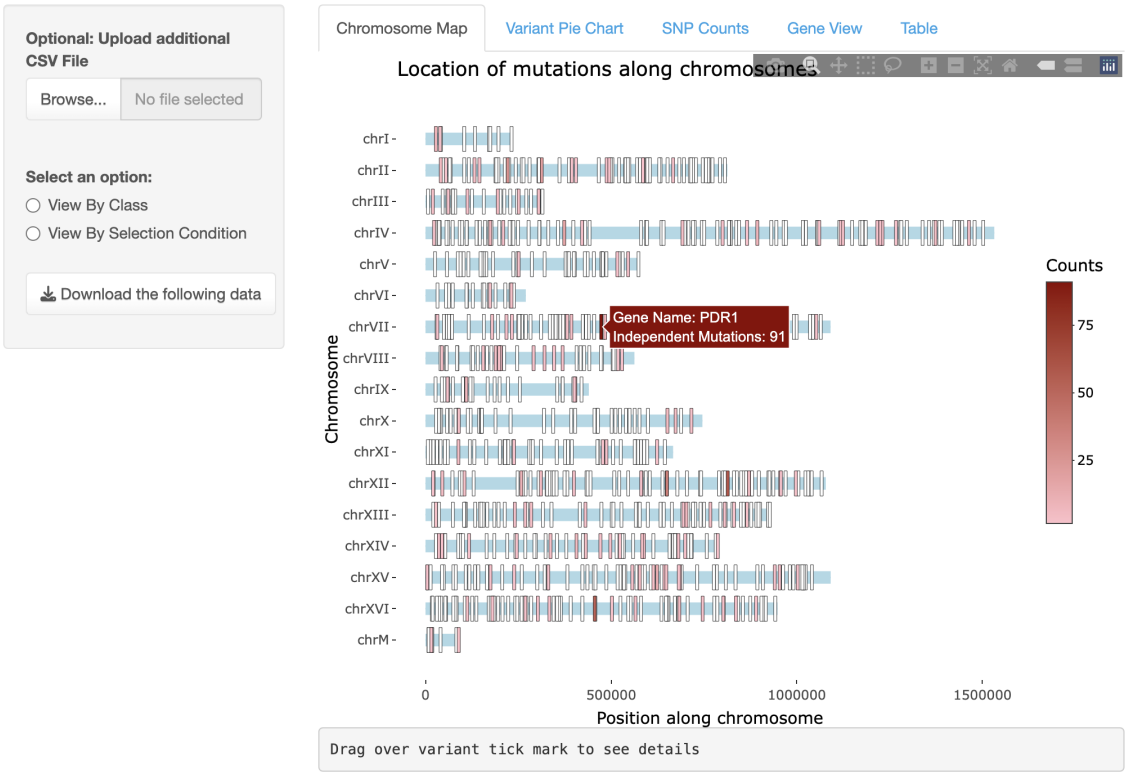

Figure 5: Unfiltered chromosome map

Variant Pie Chart

The Variant Pie Chart displays the distribution of mutation types identified by the sequencing analysis pipeline. Unlike the chromosome plot, which focuses solely on the locations of mutated genes, the pie chart encompasses all variants, including those in non-coding regions.

Default (Unfiltered) Variant Pie Chart:

SNP Counts

This plot illustrates the distribution of single nucleotide polymorphisms (SNPs) and small insertions and deletions (indels). By categorizing the different types of SNPs, the mutation spectrum provides insights into the underlying biochemical processes that drive DNA mutations.

Default (Unfiltered) SNP Counts:

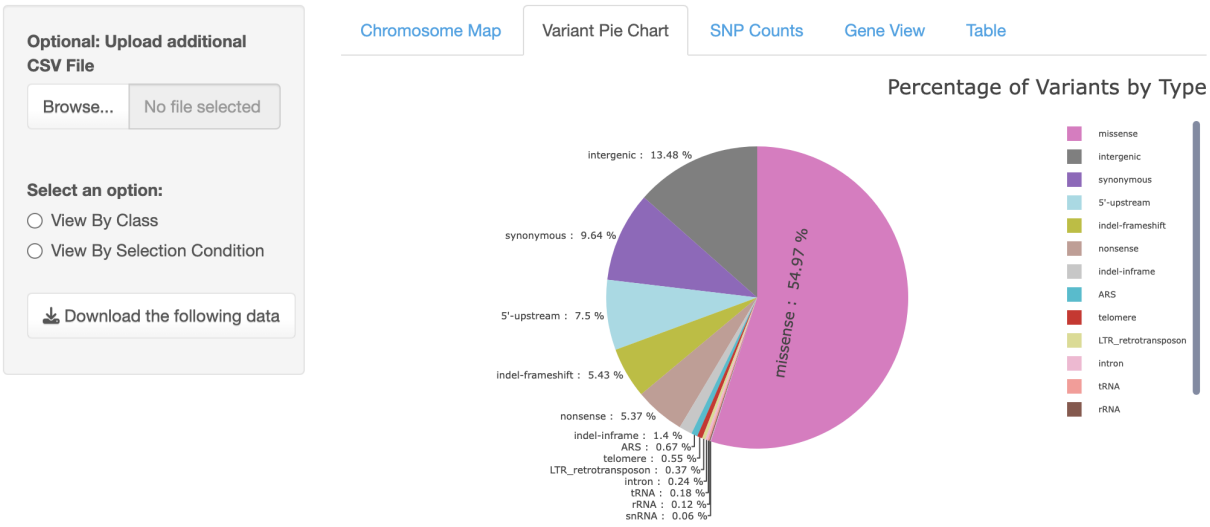

Figure 6: Default variant pie chart

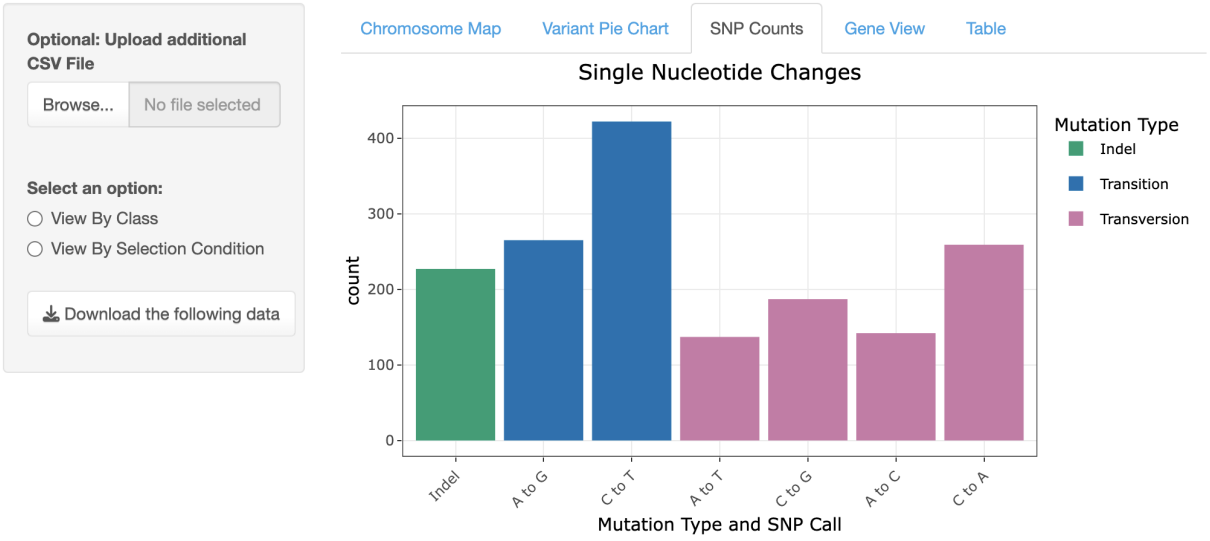

Figure 7: Default SNP counts

Gene View

Gene View is a linear lollipop plot that displays the position of coding mutations along the protein product of each mutated gene. In this view, users can select a mutated gene from a dropdown menu based on the selected dataset. Once a gene is selected, the plot reveals the length of the amino acid sequence and color-coded lollipops indicating where mutations occurred. The colors differentiate between mutation types, such as missense, nonsense, synonymous, or 5'-upstream mutations. The height of each lollipop reflects how frequently that particular site in the protein was mutated within the dataset.

Default (Unfiltered) Gene View:

Figure 8: Default gene view

Code Installation

Otherwise, download the code from this repository, navigate it via the terminal and run `Rscript app.R`.

Then, it will tell you how to access it locally by saying something such as

`Listening on http://xxxxxxx`.

Copy and paste this link into a web browser of your choice to access the mutation browser

Edit the Master VCF